2.11.2 fi ; ship specific voluntary structural enhancement
2.11.3 fi ; bulk carriers and oil tankers under Common Structural Rules (CSR)
2.11.4 fi ; other ship types
2.12 fc ; Cubic capacity correction factor
2.12.1 fc ; chemical tankers
2.12.2 fc ; LNG carriers
2.13 Lpp ; Length between perpendiculars

APPENDIX 1 A generic and simplified power plant

APPENDIX 2 Guidelines for the development of electric power tables for EEDI (EPT-EEDI)

RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
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1 Definitions

MARPOL means the International Convention for the Prevention of Pollution from

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SOURCE: MEPC.212(63).pdf
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
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I:\MEPC\63\23.doc
1 Definitions

MARPOL means the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto, as amended.

For the purpose of these Guidelines, the definitions in "REGULATIONS ON ENERGY EFFICIENCY FOR SHIPS" (RESOLUTION MEPC. 203(62)) apply.

2 Energy Efficiency Design Index (EEDI)

The attained new ship Energy Efficiency Design Index (EEDI) is a measure of ships energy efficiency (g/t*nm) and calculated by the following formula:

ref wc i
neff
i
MEFME i eff i effAEFAE
nPTI
i

neff
i
i AEeff i effi PTI
n
j
jAEFAE AEi MEi FME
nME
i
i ME
n
j
j
V f Capacity f f
SFC C P fSFC C P f P fSFC C PSFC C P f

1) ( ) (11) ( ) () (1) () (1) (1

* If part of the Normal Maximum Sea Load is provided by shaft generators, SFCME and CFME may for that part of the power be used instead of SFCAE and CFAE
** In case of P PTI(i)>0, the average weighted value of (SFCME . CFME) and

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SOURCE: MEPC.212(63).pdf
* If part of the Normal Maximum Sea Load is provided by shaft generators, SFCME and CFME may for that part of the power be used instead of SFCAE and CFAE
** In case of P PTI(i)>0, the average weighted value of (SFCME . CFME) and
(SFCAE . CFAE) to be used for calculation of Peff

Note: This formula may not be able to apply to diesel-electric propulsion, turbine propulsion or hybrid propulsion system.

Where:

.1 CF is a non-dimensional conversion factor between fuel consumption measured in g and CO 2 emission also measured in g based on carbon content. The subscripts MEi and AEi refer to the main and auxiliary engine(s) respectively. CF corresponds to the fuel used when determining SFC listed in the applicable test report included in a Technical File as defined in paragraph 1.3.15 of NO x Technical Code ("test report included in a NO x technical file" hereafter). The value of CF is as follows:

Type of fuel Reference Carbon content CF (t-CO2/t-Fuel)
1 Diesel/Gas Oil ISO 8217 Grades DMX through DMB 0.8744 3.206
2 Light Fuel Oil (LFO) ISO 8217 Grades RMA through RMD 0.8594 3.151
3 Heavy Fuel Oil
(HFO)
ISO 8217 Grades RME through

---CHUNK---

SOURCE: MEPC.212(63).pdf
Type of fuel Reference Carbon content CF (t-CO2/t-Fuel)
1 Diesel/Gas Oil ISO 8217 Grades DMX through DMB 0.8744 3.206
2 Light Fuel Oil (LFO) ISO 8217 Grades RMA through RMD 0.8594 3.151
3 Heavy Fuel Oil
(HFO)
ISO 8217 Grades RME through
RMK 0.8493 3.114
4 Liquefied Petroleum
Gas (LPG)
Propane 0.8182 3.000
Butane 0.8264 3.030
5 Liquefied Natural
Gas (LNG) 0.7500 2.750
RESOLUTION MEPC.212(63)
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2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
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.2 Vref is the ship speed, measured in nautical miles per hour (knot), on deep water in the condition corresponding to the Capacity as defined in paragraphs 2.3.1 and 2.3.3 (in case of passenger ships and ro-ro passenger ships, this condition should be summer load draught as provided in paragraph 2.4) at the shaft power of the engine(s) as defined in paragraph 2.5 and assuming the weather is calm with no wind and no waves.

.3 Capacity is defined as follows:

.1 For bulk carriers, tankers, gas tankers, ro-ro cargo ships, general

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SOURCE: MEPC.212(63).pdf
paragraph 2.5 and assuming the weather is calm with no wind and no waves.

.3 Capacity is defined as follows:

.1 For bulk carriers, tankers, gas tankers, ro-ro cargo ships, general cargo ships, refrigerated cargo carrier and combination carriers, deadweight should be used as Capacity.

.2 For passenger ships and ro-ro passenger ships, gross tonnage in accordance with the International Convention of Tonnage Measurement of Ships 1969, Annex I, regulation 3 should be used as Capacity.

.3 For containerships, 70 per cent of the deadweight (DWT) should be used as Capacity. EEDI values for containerships are calculated as follows:

.1 attained EEDI is calculated in accordance with the EEDI

formula using 70 per cent deadweight for Capacity.

.2 estimated index value in the Guidelines for calculation of the reference line is calculated using 70 per cent deadweight as:

ref
NME
i
AEi ME
V
PP
Value IndexEstimated

DWT % 70
215190
1144 . 3 1

.3 parameters a and c for containerships in Table 2 of regulation 21 of MARPOL Annex VI are determined by